Chat Link: https://parley.mit.edu/share/YU9P-ob5dZtOrfJ_9R757

```
global_data = {
    # --- Attribute Names ---
    "attribute_1": "total_cost",
    "attribute_2": "material_continuity",
    "attribute_3": "electrical_conductivity",
    "attribute_4": "mass",
    "attribute_5": "connection_reliability",

    # --- Optimization Directions ---
    "direction_attribute_1": "negative",   # Lower cost is better
    "direction_attribute_2": "positive",   # Higher continuity with incumbent is better
    "direction_attribute_3": "positive",   # Higher conductivity is better
    "direction_attribute_4": "negative",   # Lower mass is better
    "direction_attribute_5": "positive",   # Higher reliability score is better

    # --- Robust Global Min/Max ---
    # Total cost ($/unit): covers tiny low-cost Al busbars → large premium Cu busbars
    # + accommodates significant commodity price swings over time
    "attribute_1_min": 0.50,
    "attribute_1_max": 20.00,

    # Material continuity: binary by definition, natural bounds
    "attribute_2_min": 0,
    "attribute_2_max": 1,

    # Electrical conductivity (MS/m): poor Al alloy → silver-bearing / OFHC Cu
    "attribute_3_min": 15,
    "attribute_3_max": 65,

    # Mass (kg): small signal busbar (~10g) → large truck main distribution busbar (~1kg)
    "attribute_4_min": 0.010,
    "attribute_4_max": 1.000,

    # Connection reliability (0–10): full defined scale, future-proofs for
    # novel materials/joining methods scoring outside current Cu/Al range
    "attribute_5_min": 0,
    "attribute_5_max": 10,

    # --- Commodity Prices ---
    "copper_price_per_ton": 9500,
    "aluminum_price_per_ton": 2500,
  "copper_price_per_kg": 9.500,     # fixed global Cu price
```

```
    "aluminum_price_per_kg": 2.500,   # fixed global Al price
}

regional_data = [
    {
        # Strong Cu incumbency in Tier 1 supply chain (Aptiv, Lear, Delphi)
        # Truck/SUV-heavy fleet reduces mass urgency vs. Europe
        # Cold-climate cranking makes conductivity a real functional concern
        "region": "United States",
        "copper_product_market_share": 0.85,
        "aluminum_product_market_share": 0.15,
        "weight_attribute_1": 0.25,   # total_cost
        "weight_attribute_2": 0.25,   # material_continuity — HIGHEST of all regions
        "weight_attribute_3": 0.20,   # electrical_conductivity
        "weight_attribute_4": 0.15,   # mass
        "weight_attribute_5": 0.15,   # connection_reliability
    },
    {
        # Most cost-competitive OEM landscape (BYD, SAIC, Geely, Chery)
        # World's largest Al producer — Al substitution economically attractive
        # CAFC standards creating growing but moderate mass pressure
        # Younger automotive industry = lower supply chain lock-in than US
        "region": "China",
        "copper_product_market_share": 0.70,
        "aluminum_product_market_share": 0.30,
        "weight_attribute_1": 0.30,   # total_cost — elevated due to intense OEM price competition
        "weight_attribute_2": 0.15,   # material_continuity — lower; newer industry, less lock-in
        "weight_attribute_3": 0.20,   # electrical_conductivity
        "weight_attribute_4": 0.20,   # mass — moderate CAFC pressure
        "weight_attribute_5": 0.15,   # connection_reliability
    },
    {
        # Most price-sensitive market globally — cost dominates
        # Average vehicle price fraction of US/EU levels
        # Lower incumbency lock-in; pragmatic, cost-driven Tier 1 ecosystem
        # Harsh climate (heat, humidity, vibration) elevates reliability concern
        "region": "India",
        "copper_product_market_share": 0.60,
        "aluminum_product_market_share": 0.40,
        "weight_attribute_1": 0.40,   # total_cost — DOMINANT; highest of all regions
        "weight_attribute_2": 0.10,   # material_continuity — LOWEST; pragmatic switching culture
        "weight_attribute_3": 0.20,   # electrical_conductivity
        "weight_attribute_4": 0.10,   # mass — LOWEST; CAFE-equivalent standards less stringent
        "weight_attribute_5": 0.20,   # connection_reliability — elevated; extreme heat/humidity/vi
```

```
    },
    {
        # Strictest CO₂ fleet targets globally (95 g/km → 0 by 2035)
        # OEMs assign internal shadow price of €5–20/kg saved
        # Premium OEM landscape (BMW, Mercedes, VW Group) tolerates cost premiums
        # Highest long-life/durability requirements; 15+ year vehicle expectations
        "region": "Europe",
        "copper_product_market_share": 0.75,
        "aluminum_product_market_share": 0.25,
        "weight_attribute_1": 0.20,   # total_cost — LOWEST; premium OEMs less
        "weight_attribute_2": 0.15,   # material_continuity
        "weight_attribute_3": 0.20,   # electrical_conductivity
        "weight_attribute_4": 0.25,   # mass — HIGHEST; strongest CO₂ regulatory driver globally
        "weight_attribute_5": 0.20,   # connection_reliability — elevated; 15+ year life requirements
    },
]

product_data = [

    # ── UNITED STATES

    {
        # Cu PDC Busbar: ETP copper (C11000), tin-plated, stamped & bent
        # NMC includes: stamping, tin plating, terminal prep, QC, labor/overhead
        "region": "United States",
        "dominant material": "cu",
        "product": "ICE PDC Busbar",
        "copper_kg": 0.145,            # ETP copper bar body
        "aluminum_kg": 0.000,          # No aluminum content
        "total_mass_kg": 0.150,        # +0.005 kg: tin plating + minor hardware
        "non_material_cost_per_unit": 3.50,   # USD — high US labor/overhead
        "attribute_2_value": 1,        # material_continuity: Cu incumbent
        "attribute_3_value": 58,       # electrical_conductivity: 58 MS/m (ETP Cu)
        "attribute_4_value": 0.150,    # mass: 0.150 kg
        "attribute_5_value": 9,        # connection_reliability: 9/10 — no creep, stable oxide
    },
    {
        # Al PDC Busbar: 1350/6101 Al alloy, with Cu bi-metallic terminal inserts
        # NMC premium over Cu: special terminal treatment, anti-fretting compound,
        # torque management, anodizing/coating — all add ~$0.50/unit vs Cu
        "region": "United States",
        "dominant material": "al",
        "product": "ICE PDC Busbar",
        "copper_kg": 0.005,            # Cu bi-metallic inserts at terminal points
```

```
    "aluminum_kg": 0.070,          # Al conductor body (1.66× cross-section vs Cu)
    "total_mass_kg": 0.075,        # ~50% lighter than Cu equivalent
    "non_material_cost_per_unit": 4.00,   # USD — +$0.50 vs Cu for special processing
    "attribute_2_value": 0,        # material_continuity: Al incumbent
    "attribute_3_value": 35,       # electrical_conductivity: 35 MS/m (Al 1350 alloy)
    "attribute_4_value": 0.075,    # mass: 0.075 kg
    "attribute_5_value": 6,        # connection_reliability: 6/10 — creep, oxide formation
},

# ── CHINA

{
    "region": "China",
    "dominant material": "cu",
    "product": "ICE PDC Busbar",
    "copper_kg": 0.145,
    "aluminum_kg": 0.000,
    "total_mass_kg": 0.150,
    "non_material_cost_per_unit": 2.00,   # USD — lower labor/overhead vs US/Europe
    "attribute_2_value": 1,
    "attribute_3_value": 58,
    "attribute_4_value": 0.150,
    "attribute_5_value": 9,
},
{
    "region": "China",
    "dominant material": "al",
    "product": "ICE PDC Busbar",
    "copper_kg": 0.005,
    "aluminum_kg": 0.070,
    "total_mass_kg": 0.075,
    "non_material_cost_per_unit": 2.50,   # USD — +$0.50 vs Cu for special processing
    "attribute_2_value": 0,
    "attribute_3_value": 35,
    "attribute_4_value": 0.075,
    "attribute_5_value": 6,
},

# ── INDIA

{
    "region": "India",
    "dominant material": "cu",
    "product": "ICE PDC Busbar",
```

```
    "copper_kg": 0.145,
    "aluminum_kg": 0.000,
    "total_mass_kg": 0.150,
    "non_material_cost_per_unit": 1.80,   # USD — lowest labor costs of all four regions
    "attribute_2_value": 1,
    "attribute_3_value": 58,
    "attribute_4_value": 0.150,
    "attribute_5_value": 9,
},
{
    "region": "India",
    "dominant material": "al",
    "product": "ICE PDC Busbar",
    "copper_kg": 0.005,
    "aluminum_kg": 0.070,
    "total_mass_kg": 0.075,
    "non_material_cost_per_unit": 2.30,   # USD — +$0.50 vs Cu for special processing
    "attribute_2_value": 0,
    "attribute_3_value": 35,
    "attribute_4_value": 0.075,
    "attribute_5_value": 6,
},

# ── EUROPE

{
    "region": "Europe",
    "dominant material": "cu",
    "product": "ICE PDC Busbar",
    "copper_kg": 0.145,
    "aluminum_kg": 0.000,
    "total_mass_kg": 0.150,
    "non_material_cost_per_unit": 3.20,   # USD — high European labor; slightly below US
    "attribute_2_value": 1,
    "attribute_3_value": 58,
    "attribute_4_value": 0.150,
    "attribute_5_value": 9,
},
{
    "region": "Europe",
    "dominant material": "al",
    "product": "ICE PDC Busbar",
    "copper_kg": 0.005,
    "aluminum_kg": 0.070,
```

```
        "total_mass_kg": 0.075,
        "non_material_cost_per_unit": 3.70,   # USD — +$0.50 vs Cu for special processing
        "attribute_2_value": 0,
        "attribute_3_value": 35,
        "attribute_4_value": 0.075,
        "attribute_5_value": 6,
    },
]
```